

BCP Blockers Have Essential Hereditary Support

Run `fa_banach_001`, lane 19

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Source Problem and Context

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. It asks for a noncommutative C^* -algebra version of the commutative $C(K)$ BCP/UBCP criterion:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

The lane packet

`2311.14257_bcp_countable_capture_exact_equivalence`

proved that a nonzero normed space has BCP if and only if there is a countable set C such that every unit vector is not an exact capture direction for C . Equivalently, for every $u \in S_X$ some $c \in C$ satisfies

$$\text{dist}(c, \mathbb{F}u) < \|c\|.$$

We call such a set a BCP blocker.

The earlier packet

`2311.14257_bcp_forces_faithful_state_essential_blocker_ideal`

proved that the closed two-sided ideal generated by a BCP blocker is essential. The present packet strengthens this from the ideal level to the hereditary-support level.

Statement

Theorem 1 (essential hereditary support of a BCP blocker). *Let A be a nonzero C^* -algebra with BCP. Let*

$$C = \{c_1, c_2, \dots\} \subset A$$

be a countable BCP blocker, so for every $u \in S_A$ there is n with

$$\text{dist}(c_n, \mathbb{C}u) < \|c_n\|.$$

Define

$$h = \sum_{n=1}^{\infty} 2^{-n-2} \frac{c_n^* c_n + c_n c_n^*}{1 + \|c_n\|^2} \in A_+.$$

Then the hereditary subalgebra

$$B = \overline{hAh}$$

is essential in A ; equivalently its annihilator

$$\text{Ann}(B) = \{a \in A : aB = Ba = \{0\}\}$$

is zero.

Corollary 1. *Every C^* -algebra with BCP contains an essential σ -unital hereditary subalgebra generated canonically by a countable BCP blocker.*

Idea of the Proof

The two-sided ideal generated by a countable set can be much larger than any separable or σ -unital local object. For example, a rank-one projection in $\mathcal{B}(H)$, with H nonseparable, generates $\mathcal{K}(H)$ as an ideal.

The hereditary subalgebra generated by the blocker squares is more rigid. If it had a nonzero annihilator, choose a positive unit vector u in that annihilator. Since h dominates positive multiples of both $c_n^* c_n$ and $c_n c_n^*$, the relations $uh = hu = 0$ force

$$c_n u = u c_n = 0 \quad (n \geq 1).$$

Then c_n and u live in orthogonal corners, so

$$\|c_n - \lambda u\| = \max\{\|c_n\|, |\lambda|\} \quad (\lambda \in \mathbb{C}).$$

Thus u captures all of C , contradicting the blocker property.

Proof

Proof of Theorem 1. The series defining h converges in norm and gives a positive element of A . For every n , there is a constant $\alpha_n > 0$ such that

$$h \geq \alpha_n c_n^* c_n, \quad h \geq \alpha_n c_n c_n^*.$$

Suppose for contradiction that $B = \overline{hAh}$ is not essential. Then its annihilator contains a nonzero positive element. Choose

$$0 \leq u \in \text{Ann}(B), \quad \|u\| = 1.$$

Since $h \in B$, we have

$$uh = hu = 0.$$

For a fixed n , the domination above gives

$$0 \leq uc_n^*c_nu \leq \alpha_n^{-1}uhu = 0,$$

and hence $c_nu = 0$. Similarly,

$$0 \leq uc_nc_n^*u \leq \alpha_n^{-1}uhu = 0,$$

so $c_n^*u = 0$, equivalently $uc_n = 0$.

Therefore $c_nu = uc_n = 0$ for every n . It follows that for every $\lambda \in \mathbb{C}$,

$$(c_n - \lambda u)^*(c_n - \lambda u) = c_n^*c_n + |\lambda|^2u^2,$$

and the two positive summands are orthogonal. Hence

$$\|c_n - \lambda u\|^2 = \max\{\|c_n^*c_n\|, |\lambda|^2\|u^2\|\} = \max\{\|c_n\|^2, |\lambda|^2\}.$$

Thus

$$\text{dist}(c_n, \mathbb{C}u) = \|c_n\| \quad (n \geq 1).$$

This says that the unit vector u captures the whole blocker C , a contradiction. Therefore $\text{Ann}(B) = 0$, so B is essential. $\square$

Proof of Corollary 1. The hereditary subalgebra generated by a single positive element is σ -unital; for instance, it has a countable approximate identity obtained from continuous functional calculus of h . Apply Theorem 1. $\square$

Relation to the Remaining BCP/UBCP Wall

This theorem does not prove UBCP. Essential σ -unital hereditary support is still weaker than a countable local coefficient model. The separable-essential-ideal packet proves UBCP when the essential local object is separable, but the present $B = \bar{h}A\bar{h}$ can be nonseparable.

The theorem nevertheless sharpens the structural route. A hypothetical C^* -algebra with BCP but not UBCP cannot merely have a diffuse or remote BCP blocker: the blocker produces a concrete positive element whose hereditary support meets every nonzero ideal. The remaining question is:

Does an essential hereditary support coming from a BCP blocker contain enough countable local coefficient structure to build a uniform ball cover?

If yes, BCP implies UBCP for C^* -algebras. If no, the failure should produce one of the known countable-capture mechanisms.

Novelty and Search Bounds

This packet was produced during the second consolidation of the lane-19 arXiv:2311.14257 work. The local run indexes were searched for essential blocker ideals, separable right-support nets, hereditary support, essential hereditary subalgebras, and BCP/countable capture. The previous essential-blocker packet recorded essentiality of the ideal generated by the blocker, but not this hereditary-support strengthening.

Human Review Focus

Reviewers should check:

1. that h dominates positive multiples of both $c_n^*c_n$ and $c_nc_n^*$;
2. that a nonzero annihilator element u of $\overline{hAh}$ satisfies $uh = hu = 0$;
3. the implications $hu = uh = 0 \Rightarrow c_nu = uc_n = 0$;
4. the orthogonal positive-summand norm calculation;
5. the scope limitation: essential σ -unital hereditary support is not yet the same as UBCP.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Bruce Blackadar, *Operator Algebras: Theory of C^* -Algebras and Von Neumann Algebras*, Encyclopaedia of Mathematical Sciences 122, Springer, 2006.
- [3] Run `fa_banach_001`, line 19, *BCP and Countable Capture Are Exact Opposites*, solution packet 2311.14257_bcp_countable_capture_exact_equivalence, 2026.
- [4] Run `fa_banach_001`, line 19, *BCP Forces a Faithful State and an Essential Blocker Ideal*, solution packet 2311.14257_bcp_forces_faithful_state_essential_blocker_ideal, 2026.